Lecture 6

Representations of $\mathfrak{sl}_2$

Objectives. *We classify the finite-dimensional irreducible $\mathfrak{sl}_2$-modules. There is exactly one up to isomorphism for each nonnegative integer. Weight counting then decomposes tensor products.*

By the end of this lecture you will be able to:

- Write the action of h, e and f on V_m in the basis $f^k v_0$.
- Prove that every finite-dimensional irreducible $\mathfrak{sl}_2$-module is isomorphic to exactly one V_m.
- Derive the scalar by which the Casimir operator acts on V_m.
- Decompose $V_m \otimes V_n$ by weight counting and locate its highest vectors.

Throughout, $\mathfrak{sl}_2 = \mathfrak{sl}_2(\mathbb{C})$ has the basis (h, e, f) of Lecture 1, with

$$[h, e] = 2e, \qquad [h, f] = -2f, \qquad [e, f] = h.$$

Modules are finite-dimensional over $\mathbb{C}$ unless a statement says otherwise. For a module (M, ρ) we write xu for $\rho(x)u$ inside calculations. The operator is still $\rho(x)$, and $D(x)$ is its matrix in a named basis.

6.1 Weights and the modules V_m

Definition 6.1 (Weight space). Let M be an $\mathfrak{sl}_2$-module, possibly infinite-dimensional, and let $\mu \in \mathbb{C}$. The *weight space* of weight μ is

$$M[\mu] = \{u \in M : hu = \mu u\}.$$

If $M[\mu] \neq 0$, then μ is a *weight* of M. Nonzero elements of $M[\mu]$ are *weight vectors*.

Lemma 6.2 (Weight shifts). *For every module M, finite-dimensional or not, and every $\mu \in \mathbb{C}$,*

$$e\,M[\mu] \subseteq M[\mu + 2], \qquad f\,M[\mu] \subseteq M[\mu - 2].$$

Proof. Let $u \in M[\mu]$. The relations $[h, e] = 2e$ and $[h, f] = -2f$ give

$$h(eu) = e(hu) + 2eu = (\mu + 2)\,eu,$$
$$h(fu) = f(hu) - 2fu = (\mu - 2)\,fu.$$

■

Definition 6.3 (The module V_m). Let $m \geq 0$ be an integer. Let V_m have basis $v_0, \dots, v_m$, and put $v_{-1} = v_{m+1} = 0$. For $0 \leq k \leq m$ define linear operators by

$$hv_k = (m - 2k)v_k, \qquad fv_k = v_{k+1}, \qquad ev_k = k(m - k + 1)v_{k-1}.$$

Proposition 6.4 (The modules V_m · *P*)**.** *The formulas define a representation of $\mathfrak{sl}_2$ on V_m. The module V_m is irreducible of dimension $m+1$. Its weights are $m, m-2, \dots, -m$, and each weight space is one-dimensional.*

Proof. It suffices to check the three relations on each v_k. For $[e,f]=h$,

$$\begin{aligned}(ef-fe)v_k &= ev_{k+1} - k(m-k+1)\,fv_{k-1}\\ &= \big((k+1)(m-k) - k(m-k+1)\big)v_k = (m-2k)v_k.\end{aligned}$$

At $k=m$ the term $ev_{m+1}=0$ matches the factor $m-k=0$. At $k=0$ the second term carries the factor $k=0$.

The vector ev_k is a multiple of v_{k-1}, and $fv_k = v_{k+1}$. Hence

$$\begin{aligned}(he-eh)v_k &= k(m-k+1)\big((m-2k+2)-(m-2k)\big)v_{k-1} = 2\,ev_k,\\ (hf-fh)v_k &= \big((m-2k-2)-(m-2k)\big)v_{k+1} = -2\,fv_k.\end{aligned}$$

The weights $m-2k$ are read from the formula for h. They are distinct, so each weight space is one-dimensional.

For irreducibility, let $W\neq 0$ be a submodule and let $w=\sum_k c_k v_k \in W$ be nonzero. Choose the Lagrange interpolation polynomial p_k with $p_k(m-2j)=\delta_{jk}$ for $0\le j\le m$. Since W is stable under $\rho(h)$,

$$p_k(\rho(h))\,w = c_k v_k \in W.$$

Pick k with $c_k\neq 0$. Then $v_k\in W$ and

$$e^k v_k = \prod_{j=1}^{k} j(m-j+1)\;v_0.$$

Each factor is nonzero for $1\le j\le k\le m$. Hence $v_0\in W$, and then $v_j = f^j v_0 \in W$ for every j. Thus $W=V_m$. ∎

Example 6.5 (Small values of m)**.** The module V_0 is trivial. For V_1, take $v_0=(1,0)^{\mathsf T}$ in the defining module $\mathbb{C}^2$. Then $v_1 = fv_0 = (0,1)^{\mathsf T}$, $ev_1 = v_0$ and $ev_0 = fv_1 = 0$. The vectors v_0 and v_1 have weights 1 and -1. So V_1 is the defining module.

For V_2 in the ordered basis (v_0, v_1, v_2), the action matrices are

$$D(h)=\operatorname{diag}(2,0,-2),\qquad D(e)=\begin{pmatrix}0&2&0\\0&0&2\\0&0&0\end{pmatrix},\qquad D(f)=\begin{pmatrix}0&0&0\\1&0&0\\0&1&0\end{pmatrix}.$$

The adjoint module realises V_2. Put $v_0 = e$. Then

$$v_1 = [f,e] = -h,\qquad v_2 = [f,-h] = -2f,\qquad [f,v_2]=0.$$

The brackets with h give v_0, v_1 and v_2 the weights 2, 0 and -2. The action of e is as required:

$$[e,v_0]=0,\qquad [e,v_1]=[h,e]=2v_0,\qquad [e,v_2]=-2h=2v_1.$$

6.2 Highest vectors and the classification

Definition 6.6 (Highest vector)**.** Let M be a module, possibly infinite-dimensional, and let $\lambda \in \mathbb{C}$. A *highest vector* of weight λ is a nonzero $u \in M[\lambda]$ with $eu = 0$.

In V_m the vector v_0 is a highest vector of weight m.

Lemma 6.7 (The f-string · *P*)**.** *Let u be a highest vector of weight λ in a module M, possibly infinite-dimensional. Put $u_k = f^k u$ for $k \geq 0$ and $u_{-1} = 0$. Then*

$$hu_k = (\lambda - 2k)u_k, \qquad fu_k = u_{k+1}, \qquad eu_k = k(\lambda - k + 1)\, u_{k-1}.$$

Proof. The first formula is Lemma 6.2 applied k times. The second is the definition. The third holds at $k = 0$ because $eu = 0$. Assume the third formula for $k - 1 \geq 0$. Since $ef = fe + h$ as operators,

$$\begin{aligned} eu_k &= f\,(eu_{k-1}) + hu_{k-1} \\ &= \big((k-1)(\lambda - k + 2) + (\lambda - 2k + 2)\big)u_{k-1} = k(\lambda - k + 1)\, u_{k-1}. \end{aligned}$$

Both coefficients in the last step expand to $k\lambda - k^2 + k$. ∎

Theorem 6.8 (Highest vectors in finite dimension · *P*)**.** *Let M be finite-dimensional and let $u \in M$ be a highest vector of weight λ. Put $u_k = f^k u$ for $k \geq 0$. Then:*

(i) *$\lambda = m$ for some integer $m \geq 0$;*

(ii) *$u_0, \dots, u_m$ are linearly independent and $u_{m+1} = 0$;*

(iii) *the linear map $\iota\colon V_m \to M$ with $\iota(v_k) = u_k$ is an injective intertwiner.*

Proof. By Lemma 6.7, each nonzero u_k is an eigenvector of $\rho(h)$ with eigenvalue $\lambda - 2k$. These eigenvalues are distinct. Hence the nonzero vectors u_k are linearly independent. As $\dim M$ is finite, some u_k vanishes. Let $m+1$ be the least such index. Since $u_0 = u \neq 0$, we have $m \geq 0$. The vectors $u_0, \dots, u_m$ are nonzero, hence independent. Also $u_k = f^{k-m-1}u_{m+1} = 0$ for every $k > m$.

Apply the third formula of Lemma 6.7 with $k = m + 1$:

$$0 = eu_{m+1} = (m+1)(\lambda - m)\, u_m.$$

Since $m + 1 \neq 0$ and $u_m \neq 0$, we get $\lambda = m$. This proves (i) and (ii).

With $\lambda = m$, the formulas of Lemma 6.7 for $0 \leq k \leq m$ are the formulas defining V_m. This includes $fu_m = u_{m+1} = 0$. Thus

$$\iota(xv_k) = x\,\iota(v_k) \qquad \text{for } x = h, e, f \text{ and } 0 \leq k \leq m.$$

By linearity in x and in the vector, ι is an intertwiner. It is injective by (ii). ∎

In a finite-dimensional module the f-string must stop. Stopping requires the coefficient $(m+1)(\lambda - m)$ to vanish.

Without finite dimensionality the string need not stop. Exercise 6.7 studies such an infinite-dimensional module, and Lecture 12 develops the theory of these modules.

Lemma 6.9 (Existence of highest vectors · *P*)**.** *Every nonzero finite-dimensional module M contains a highest vector.*

Proof. Since $\mathbb{C}$ is algebraically closed, $\rho(h)$ has an eigenvector w, of some weight μ. By Lemma 6.2, each nonzero $e^j w$ has weight $\mu + 2j$. Vectors of distinct weights are linearly independent, and $\dim M$ is finite. Hence $e^j w = 0$ for some $j \geq 1$. If j is the least such exponent, then $e^{j-1}w$ is a highest vector. ∎

Theorem 6.10 (Classification · *P*)**.** *Every finite-dimensional irreducible $\mathfrak{sl}_2$-module V is isomorphic to V_m for exactly one integer $m \geq 0$, namely $m = \dim V - 1$.*

Proof. By Lemma 6.9, V has a highest vector. By Theorem 6.8, its weight is an integer $m \geq 0$, and there is an injective intertwiner $\iota \colon V_m \to V$. The image of ι is a nonzero submodule of V. Since V is irreducible, this image is V. Hence ι is an isomorphism. Since $\dim V_m = m + 1$, the integer m is determined by V. ∎

Lecture 4 showed that the Killing form of $\mathfrak{sl}_2$ is nondegenerate, so $\mathfrak{sl}_2$ is semisimple. Hence Weyl's theorem from Lecture 5 applies. Every finite-dimensional $\mathfrak{sl}_2$-module is a direct sum of irreducible submodules. This extends the classification to all finite-dimensional modules.

Corollary 6.11 (Multiplicities from weights · *P*)**.** *Let M be a finite-dimensional $\mathfrak{sl}_2$-module. Then $\rho(h)$ is diagonalisable with integer eigenvalues, and $\dim M[\mu] = \dim M[-\mu]$ for all μ. In every decomposition of M into irreducible submodules, the number of summands isomorphic to V_μ, for an integer $\mu \geq 0$, is*

$$\dim M[\mu] - \dim M[\mu + 2].$$

Proof. Take any decomposition $M = M_1 \oplus \cdots \oplus M_r$ into irreducible submodules, which exists by Weyl's theorem. By Theorem 6.10, $M_i \cong V_{m_i}$. Each M_i is $\rho(h)$-stable, so

$$M[\mu] = \bigoplus_{i=1}^{r} M_i[\mu].$$

By Proposition 6.4, $M_i[\mu]$ is one-dimensional when μ is an integer with $|\mu| \leq m_i$ and $\mu \equiv m_i \pmod 2$. Otherwise it is zero. Hence $\rho(h)$ is diagonalisable with integer eigenvalues. For integer μ,

$$\dim M[\mu] = \#\{\, i : m_i \geq |\mu|,\ m_i \equiv \mu \pmod 2 \,\}.$$

For noninteger μ, $M[\mu] = 0$. The count is unchanged by $\mu \mapsto -\mu$. For integer $\mu \geq 0$, the summands counted at μ but not at $\mu + 2$ are exactly those with $m_i = \mu$. ∎

6.3 The Casimir scalar

Use the trace form of the defining module,

$$B(x, y) = \operatorname{tr}(xy), \qquad x, y \in \mathfrak{sl}_2.$$

It is invariant by Lecture 4. Its nonzero values on the basis are

$$B(h, h) = 2, \qquad B(e, f) = B(f, e) = 1.$$

Hence B is nondegenerate, and the B-dual basis of (h, e, f) is $(\tfrac{1}{2}h, f, e)$. This fixes the normalisation for $\mathfrak{sl}_2$.

The functional α on $\mathbb{C}h$ with $\alpha(h) = 2$ corresponds under B to h itself, so its squared length is $B(h,h) = 2$. In Lecture 7, α becomes the root of $\mathfrak{sl}_2$. The Killing form is $K = 4B$ by Lecture 4.

For a module (M, ρ), the Casimir operator of B is

$$\Omega_M = \frac{1}{2}\rho(h)^2 + \rho(e)\rho(f) + \rho(f)\rho(e).$$

Proposition 6.12 (The Casimir scalar · *P*)**.** *The operator Ω_M commutes with $\rho(x)$ for every $x \in \mathfrak{sl}_2$. On V_m it is the scalar*

$$\Omega_{V_m} = \frac{m(m+2)}{2}\,\mathrm{id}\,.$$

Proof. Lecture 5 proved that, for any invariant nondegenerate form on $\mathfrak{g}$ and any representation ρ, the operator $\sum_i \rho(x_i)\rho(x^i)$ built from a basis and its dual basis commutes with every $\rho(x)$. Applied to B and the dual bases above, this is the first claim.

Since V_m is irreducible, Schur's lemma (Lecture 1) shows that Ω_{V_m} is a scalar. To find this scalar, use $ef = fe + h$ as operators:

$$\Omega_{V_m} = \frac{1}{2}h^2 + h + 2fe.$$

Since $ev_0 = 0$ and $hv_0 = mv_0$,

$$\Omega_{V_m} v_0 = \left(\frac{m^2}{2} + m\right)v_0 = \frac{m(m+2)}{2}\,v_0.$$ ∎

On the defining module the matrices give

$$D(h)^2 = I_2, \qquad D(e)D(f) + D(f)D(e) = I_2.$$

So $\Omega_{V_1} = \frac{3}{2}\,\mathrm{id}$, as the formula gives for $m = 1$. The scalars $m(m+2)/2$ strictly increase with m, so the Casimir scalar distinguishes the modules V_m.

Since $K = 4B$, the K-dual basis of (h, e, f) is $(\frac{1}{8}h, \frac{1}{4}f, \frac{1}{4}e)$. Hence the Casimir operator of K is $\Omega/4$, with scalar $m(m+2)/8$ on V_m. On the adjoint module V_2 this scalar is 1.

6.4 Tensor products and the Clebsch–Gordan rule

For modules (V, ρ_V) and (W, ρ_W), define the tensor action by

$$\rho(x) = \rho_V(x) \otimes \mathrm{id}_W + \mathrm{id}_V \otimes \rho_W(x).$$

Operators on different factors commute. The two same-factor commutators give

$$[\rho(x), \rho(y)] = \rho_V([x,y]) \otimes \mathrm{id}_W + \mathrm{id}_V \otimes \rho_W([x,y]) = \rho([x,y]).$$

Thus the tensor action is a representation. If $v \in V[\mu]$ and $w \in W[\nu]$, then

$$h(v \otimes w) = hv \otimes w + v \otimes hw = (\mu + \nu)\, v \otimes w.$$

Let $v_0, \dots, v_m$ and $w_0, \dots, w_n$ be the standard bases of V_m and V_n. The vectors $v_i \otimes w_j$ form a basis of weight vectors. The weight of $v_i \otimes w_j$ is

$$(m - 2i) + (n - 2j) = m + n - 2(i + j).$$

For $0 \le s \le m + n$ write

$$d(s) = \dim(V_m \otimes V_n)[m + n - 2s] = \#\{(i,j) : 0 \le i \le m,\ 0 \le j \le n,\ i + j = s\},$$

and put $d(-1) = 0$.

Theorem 6.13 (Clebsch–Gordan rule · *P*). *For integers $m, n \geq 0$,*

$$V_m \otimes V_n \cong \bigoplus_{k=0}^{\min(m,n)} V_{m+n-2k}.$$

Proof. First we count weight vectors. For fixed s, the pair $(i, s-i)$ is admissible exactly when

$$\max(0, s-n) \leq i \leq \min(m, s).$$

We claim that

$$d(s) = \min(s, m, n) + 1 \quad \text{for } 0 \leq s \leq \frac{m+n}{2}.$$

If $s \leq \min(m,n)$, then i runs from 0 to s. If $n < s$, then $n < m$ and $s \leq (m+n)/2 < m$, so i runs from $s-n$ to s, giving $n+1$ values. If $m < s$, the same argument with m and n exchanged gives $m+1$ values.

Now apply Corollary 6.11 to $M = V_m \otimes V_n$. Let $\mu \geq 0$ be an integer. If $\mu > m+n$ or $\mu \not\equiv m+n \pmod 2$, then $M[\mu] = 0$, so V_μ does not occur. Otherwise $\mu = m+n-2s$ with $0 \leq s \leq (m+n)/2$, and $\mu+2$ corresponds to $s-1$. The multiplicity of V_μ is

$$d(s) - d(s-1) = \begin{cases} 1, & 0 \leq s \leq \min(m,n), \\ 0, & \min(m,n) < s \leq (m+n)/2. \end{cases}$$

Writing $k = s$ gives the stated decomposition. ∎

As a check, put $p = \min(m,n)$. The dimensions agree:

$$\sum_{k=0}^{p} (m+n-2k+1) = (p+1)(m+n+1-p) = (m+1)(n+1).$$

For example, $V_1 \otimes V_1 \cong V_2 \oplus V_0$ and $V_2 \otimes V_2 \cong V_4 \oplus V_2 \oplus V_0$, with $9 = 5+3+1$.

The proof finds multiplicities, not subspaces. To locate a summand, find a highest vector, that is, a weight vector u with $eu = 0$. Theorem 6.8 then gives an injective intertwiner whose image contains u.

6.5 Worked example: $V_2 \otimes V_1$

Let v_0, v_1, v_2 and w_0, w_1 be the standard bases of V_2 and V_1. The nonzero actions of e and f are

$$fv_0 = v_1, \quad fv_1 = v_2, \quad ev_1 = 2v_0, \quad ev_2 = 2v_1,$$
$$fw_0 = w_1, \quad ew_1 = w_0.$$

The weight vectors $v_i \otimes w_j$ are grouped as follows:

weight	basis of the weight space
3	$v_0 \otimes w_0$
1	$v_1 \otimes w_0,\ v_0 \otimes w_1$
-1	$v_2 \otimes w_0,\ v_1 \otimes w_1$
-3	$v_2 \otimes w_1$

By Corollary 6.11, V_3 occurs $1-0=1$ time and V_1 occurs $2-1=1$ time. Hence

$$V_2 \otimes V_1 \cong V_3 \oplus V_1, \qquad 6 = 4+2.$$

The summand V_3. The vector $u = v_0 \otimes w_0$ has weight 3, and $eu = ev_0 \otimes w_0 + v_0 \otimes ew_0 = 0$. A representative step of its f-string is

$$\begin{aligned} f(v_1 \otimes w_0 + v_0 \otimes w_1) &= fv_1 \otimes w_0 + v_1 \otimes fw_0 + fv_0 \otimes w_1 + v_0 \otimes fw_1 \\ &= v_2 \otimes w_0 + 2\, v_1 \otimes w_1. \end{aligned}$$

Let $v_0', \dots, v_3'$ be the basis of V_3. The inclusion $\iota_3(v_k') = f^k u$ of Theorem 6.8 is

$$\begin{aligned} \iota_3(v_0') &= v_0 \otimes w_0, \\ \iota_3(v_1') &= v_1 \otimes w_0 + v_0 \otimes w_1, \\ \iota_3(v_2') &= v_2 \otimes w_0 + 2\, v_1 \otimes w_1, \\ \iota_3(v_3') &= 3\, v_2 \otimes w_1, \end{aligned}$$

and $f^4 u = 0$.

The summand V_1. A vector of weight 1 has the form $x = a\, v_1 \otimes w_0 + b\, v_0 \otimes w_1$. Then

$$ex = 2a\, v_0 \otimes w_0 + b\, v_0 \otimes w_0 = (2a + b)\, v_0 \otimes w_0.$$

So $ex = 0$ exactly when $b = -2a$. Take

$$x = v_1 \otimes w_0 - 2\, v_0 \otimes w_1.$$

Let v_0'', v_1'' be the basis of V_1. The inclusion $\iota_1(v_k'') = f^k x$ is

$$\begin{aligned} \iota_1(v_0'') &= v_1 \otimes w_0 - 2\, v_0 \otimes w_1, \\ \iota_1(v_1'') &= v_2 \otimes w_0 + v_1 \otimes w_1 - 2\, v_1 \otimes w_1 \\ &= v_2 \otimes w_0 - v_1 \otimes w_1. \end{aligned}$$

The next term vanishes, as Theorem 6.8 predicts:

$$f^2 x = v_2 \otimes w_1 - v_2 \otimes w_1 = 0.$$

The sum is direct. In weight 1, use coordinates for $(v_1 \otimes w_0,\ v_0 \otimes w_1)$. In weight -1, use coordinates for $(v_2 \otimes w_0,\ v_1 \otimes w_1)$. In each matrix below, the first row is the coordinate vector of the ι_3-image and the second row that of the ι_1-image:

$$\det \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix} = -3, \qquad \det \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix} = -3.$$

The weights ± 3 are covered by ι_3 alone. So the six image vectors form a basis, and

$$V_2 \otimes V_1 = \iota_3(V_3) \oplus \iota_1(V_1)$$

as an internal direct sum of submodules. By Proposition 6.12, Ω acts by $\frac{15}{2}$ and $\frac{3}{2}$ on the two summands.

Self-checks

(1) Can a finite-dimensional $\mathfrak{sl}_2$-module contain a vector $u \neq 0$ with $hu = -2u$ and $eu = 0$?

(2) Decompose $V_3 \otimes V_2$ and check the dimensions.

(3) A module M has $\dim M[\mu] = 1, 2, 3, 2, 1$ for $\mu = 2, 1, 0, -1, -2$, and no other weights. Decompose M and list the eigenvalues of Ω_M.

Answers. (1) No. Such a u is a highest vector, and Theorem 6.8 forces its weight to be a nonnegative integer. Vectors of weight -2 do occur, for example $v_2 \in V_2$, but $ev_2 = 2v_1 \neq 0$. (2) $V_3 \otimes V_2 \cong V_5 \oplus V_3 \oplus V_1$, with $12 = 6 + 4 + 2$. (3) The multiplicities are $1 - 0 = 1$ for V_2, $2 - 0 = 2$ for V_1 and $3 - 1 = 2$ for V_0. So $M \cong V_2 \oplus V_1^{\oplus 2} \oplus V_0^{\oplus 2}$, of dimension 9. The eigenvalues of Ω_M are 4, $\frac{3}{2}$ and 0.

Summary

The operators $\rho(e)$ and $\rho(f)$ change weights by 2 and -2. In a finite-dimensional module, the f-string of a highest vector of weight λ must stop. If it has $m+1$ nonzero vectors, the coefficient $(m+1)(\lambda-m)$ forces $\lambda = m \in \{0, 1, 2, \dots\}$, and the string spans a copy of V_m. Hence every finite-dimensional irreducible module is isomorphic to exactly one V_m.

By Weyl's theorem, weight dimensions determine every finite-dimensional module up to isomorphism. This gives the Clebsch–Gordan rule. The Casimir operator acts on V_m by $m(m+2)/2$. Lecture 8 applies these results to the subalgebras attached to roots.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Section 7. Erdmann–Wildon, *Introduction to Lie Algebras*, Chapter 8. Fulton–Harris, *Representation Theory: A First Course*, Lecture 11, §§11.1–11.2. Exercise Sheet 6 treats symmetric powers, invariant forms, singlets, tensor Casimir operators, further decompositions and infinite-dimensional strings.